When Neither Minimum Variance nor Hierarchical Risk Parity Is Optimal: An Analytical Result for the Schur Bridge Between Them

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Abstract

Schur-complementary allocation interpolates hierarchical risk parity ($\gamma = 0$) and minimum variance ($\gamma \to 1$) through a coupling γ that sets how much of the estimated cross-block covariance the recursion uses. We prove that under covariance estimation error neither endpoint is optimal: for all sufficiently small noise the expected out-of-sample variance strictly improves away from hierarchical risk parity, and above an explicit noise threshold it strictly deteriorates toward full coupling, so the minimizer is interior — a conclusion made unconditional by a four-asset example verified in exact rational arithmetic. Under natural shape conditions the optimal coupling is the unique root of $\tau^2 = -V_0'(\gamma)/G'(\gamma)$ and decreases monotonically in the noise level, sweeping the entire bridge: every interior coupling is optimal at exactly one noise level, minimum variance only as the noise vanishes, hierarchical risk parity only as it diverges.

Keywords: portfolio construction; hierarchical risk parity; Schur complement; estimation error; shrinkage; bias–variance trade-off.
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1 Introduction

Hierarchical risk parity [1] allocates by recursive bisection without inverting a covariance matrix; minimum variance inverts it in full. That the two are ends of a single construction — augmenting each block with a γ-scaled Schur complement of its sibling yields a recursion that is exactly HRP at $\gamma = 0$ and recovers minimum variance as $\gamma \to 1$ — is a unification that first appeared in a blog [2], then in a textbook [3, §12.3.4], with a fuller development in [4]; on the hierarchical graph structures of [6] the $\gamma = 1$ recovery is exact. The endpoints have since been compared analytically: [5] derive the weight noise of HRP and of Markowitz under covariance estimation error and prove HRP is the more robust.

This paper proves the statistical claim implicit in the bridge itself: under estimation noise the optimal coupling is generically *interior*, and its location is characterized by a bias–variance first-order condition. The contributions are four short results. (i) A structural lemma: the $\gamma = 0$ recursion never reads the top-level cross-block, so cross-block noise cannot move it. (ii) If coupling has first-order population value, HRP is strictly suboptimal under all sufficiently small noise, of any support. (iii) Because the estimated cross-block enters the recursion only with a γ prefactor,

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the noise cost vanishes quadratically at the HRP end while it is order one at full coupling; above an explicit threshold the minimizer is interior — and a witness in the minimal non-trivial configuration (four assets in two blocks, the smallest universe on which γ acts at all), verified in exact rational arithmetic, makes the conclusion unconditional. (iv) Under shape conditions verified in the witness family, the optimal coupling $\gamma^*(\tau^2)$ is the unique root of $\tau^2 = -V_0'/G'$ and decreases monotonically in the noise, sweeping the entire bridge: an optimal-shrinkage formula in the spirit of [7].

2 The recursion

The family. Following [4], fix an asset order and a balanced bisection tree; assume for notational simplicity that n is a power of two, so sibling blocks have equal size (a step-up matrix enters in the general case [4]). A hierarchical allocation assigns a node's capital to its children (L, R) in the ratio $1/\nu(\cdot_L) : 1/\nu(\cdot_R)$ for an inverse-fitness functional ν, and recurses. Writing a node's block of the covariance as $\left(\begin{smallmatrix} A & B \\ B^\top & D \end{smallmatrix}\right)$, hierarchical risk parity passes A and D to the children unchanged; the Schur-complementary family instead passes *augmented* blocks built from the γ-damped Schur complement and its companion vector,

$$A^c(\gamma) \;=\; A - \gamma\, BD^{-1}B^\top, \qquad b_A(\gamma) \;=\; \mathbf{1} - \gamma\, BD^{-1}\mathbf{1}, \tag{1}$$

and symmetrically for $D^c(\gamma), b_D(\gamma)$. In the canonical presentation the intra-group matrix is $A^c(\gamma)$ with an elementwise division by $b_A b_A^\top$ and the inter-group fitness is evaluated on $\left(A^c(\gamma)^{-1} \circ b_A b_A^\top\right)^{-1}$; at $\gamma = 0$ every member is HRP, and at $\gamma = 1$ the scheme replicates the minimum-variance portfolio exactly. For completeness we record the one-step form of that equivalence, following Appendix A of [4].

Proposition 1 ($\gamma = 1$ replicates minimum variance)**.** *For $b \neq 0$ let $w(Q, b) = \arg\min_{w: b^\top w = 1} w^\top Q w = Q^{-1}b/(b^\top Q^{-1} b)$ with attained variance $\nu(Q, b) = 1/(b^\top Q^{-1} b)$. Then*

$$\Sigma^{-1}\mathbf{1} \;=\; \begin{pmatrix} \nu(A^c, b_A)^{-1}\, w(A^c, b_A) \\ \nu(D^c, b_D)^{-1}\, w(D^c, b_D) \end{pmatrix}, \qquad A^c = A^c(1),\; b_A = b_A(1),\; \textit{etc.},$$

so the two-block scheme with these sub-portfolios and inter-group ratio $1/\nu(A^c, b_A) : 1/\nu(D^c, b_D)$ yields $w \propto \Sigma^{-1}\mathbf{1}$; recursive application to the sub-problems (each again of the form $Q^{-1}b$) extends this down the tree.

Proof. The block inversion identity gives $\Sigma^{-1}\mathbf{1} = \left((A^c)^{-1} b_A;\; (D^c)^{-1} b_D\right)$ with $A^c = A - BD^{-1}B^\top$, $b_A = \mathbf{1} - BD^{-1}\mathbf{1}$. For any SPD Q, from the displayed solution of the constrained problem, $Q^{-1}b = w(Q, b)\,(b^\top Q^{-1} b) = w(Q, b)/\nu(Q, b)$. Apply this to each block. $\square$

The implementations trade this exactness for simplicity: the collapsed matrix below satisfies $\widetilde{A}^{-1}\mathbf{1} = (A^c)^{-1} b_A$ at the vector level, but the naive fitness (3) and the symmetrization make $\gamma = 1$ *approach* the minimum-variance portfolio rather than replicate it. None of the results below rely on exactness at $\gamma = 1$.

The variant analyzed. Appendix A of [4] licenses an equivalent "collapse" of the pair (A^c, b_A) into a single matrix $\widetilde{A}$ with $\widetilde{A}^{-1}\mathbf{1} = A^c(\gamma)^{-1} b_A(\gamma)$. The reference implementations — `skfolio`'s `SchurComplementary` and the `allocation` package[1] — take exactly this route, and it is the recursion we analyze: each child block is replaced by

$$A_\gamma \;=\; \mathrm{sym}\Big[\big(I - \gamma BD^{-1}\big)^{-1} A^c(\gamma)\Big], \qquad D_\gamma \;=\; \big(A \leftrightarrow D,\; B \to B^\top\big), \tag{2}$$

[1] `github.com/skfolio/skfolio` and `github.com/microprediction/allocation`; the two use identical augmentation and fitness routines.

where $\mathrm{sym}(X) = \frac{1}{2}(X + X^\top)$; note $(I - \gamma B D^{-1})\mathbf{1} = b_A(\gamma)$, so (2) is the collapse of (1), up to the symmetrization. The fitness is the *naive portfolio variance*

$$\nu(X) \;=\; \tilde{w}^\top X\, \tilde{w}, \qquad \tilde{w}_i \;=\; \frac{1/X_{ii}}{\sum_j 1/X_{jj}}, \tag{3}$$

so the left child receives the fraction $\nu(D_\gamma)/\big(\nu(A_\gamma)+\nu(D_\gamma)\big)$ of the node's capital, and the recursion continues inside each augmented block down to singletons (nodes whose children are singletons perform no augmentation, matching the implementations, so a two-asset node splits by the inverse of its — already augmented — diagonal entries). At $\gamma = 0$ the augmentation (2) is the identity and the recursion is exactly HRP over the fixed order. Implementations optionally cap γ at an in-sample variance turning point (`keep_monotonic`); we analyze the raw recursion.

Two structural facts, used throughout, hold across the entire family and not just the variant (2): by (1) the cross-block B enters every member only through the products $\gamma B D^{-1}(\cdot)$, and at $\gamma = 0$ the augmentation is the identity with $b_A = \mathbf{1}$.

Let $\Sigma \succ 0$ be the population covariance and $\widehat{\Sigma}_\tau = \Sigma + \tau N$ an estimate, with N a bounded random symmetric perturbation on a set where every matrix in (2) remains invertible for $\gamma \in [0, \bar{\gamma}]$. The object of study is the expected out-of-sample variance and its population limit,

$$F(\gamma;\tau) \;=\; \mathbb{E}\big[\, w(\gamma;\widehat{\Sigma}_\tau)^\top \Sigma\, w(\gamma;\widehat{\Sigma}_\tau)\,\big], \qquad V_0(\gamma) \;=\; F(\gamma;0), \tag{4}$$

where $w(\gamma; M)$ denotes the allocation the recursion produces from an input matrix M. Every operation is rational in (γ, M), so w, F, V_0 are rational functions where denominators do not vanish — in particular C^∞ there, which is all the regularity used below.

3 Neither endpoint

Lemma 1 (The HRP end is blind to the cross-block)**.** *$w(0; M)$ is a function of (A, D) alone. Consequently, if the estimation error is supported on the top-level cross-block, then $w(0;\widehat{\Sigma}_\tau) = w(0;\Sigma)$ almost surely and $F(0;\tau) = V_0(0)$ exactly, for every τ.*

Proof. At $\gamma = 0$ the augmentation (2) returns its first argument, so the top-level split uses $\nu(A)$ and $\nu(D)$, which read only within-block entries, and the recursion continues inside the unmodified blocks A and D, never touching B. (Cross-blocks interior to A or D *are* read, through the naive variances of their parents; only the top-level B is never read.) □

Remark 1 (Scope)*.* The lemma is per-tree and per-variant: this recursion (naive variance over full child sub-blocks, as in [1]) never reads the top-level cross-block; a split on diagonal variances alone would read nothing off-diagonal. When the tree itself is data-driven (a dendrogram, or Fiedler seriation [4]), the ordering step does read the cross-block — but only through the discrete tree choice, which for a generic estimate is locally constant. The $\gamma = 0$ allocation is then locally constant in the cross-block entries, and the differential statements below are unaffected for noise small enough to leave the tree unchanged. (In the witness of Proposition 2 the seriation returns the same top partition on both noise branches, so fixing the order there is without loss.)

Theorem 1 (HRP is strictly suboptimal under small noise)**.** *If $V_0'(0) < 0$ — coupling has first-order population value — then there is $\tau_0 > 0$ such that for all $\tau < \tau_0$, $\partial_\gamma F(0;\tau) < 0$: every minimizer of $F(\cdot\,;\tau)$ on $[0,\bar{\gamma}]$ is bounded away from 0. No support restriction on N is needed.*

Proof. $\partial_\gamma F(0;\tau) = \mathbb{E}[\partial_\gamma V(0;\Sigma+\tau N)]$, where $V(\gamma;M) = w(\gamma;M)^\top \Sigma\, w(\gamma;M)$ is C^1 near $(0,\Sigma)$ with $\partial_\gamma V$ continuous and, for bounded N, dominated there. By dominated convergence $\partial_\gamma F(0;\tau) \to V_0'(0) < 0$ as $\tau \to 0$; continuity in τ gives τ_0. □

Theorem 2 (Full coupling is suboptimal above a noise threshold)**.** *Let the error be supported on the top-level cross-block and symmetric ($N \overset{d}{=} -N$). Then:*

(i) *there is $C < \infty$, uniform on a compact set of γ and noise realizations keeping* (2) *invertible, with $|F(\gamma;\tau) - V_0(\gamma)| \le C\gamma^2\tau^2$;*

(ii) *if for some $\hat\gamma \in (0,1)$ and τ^* the excess noise cost at full coupling satisfies $F(1;\tau^*) - V_0(1) > V_0(\hat\gamma) - V_0(1) + C\hat\gamma^2\tau^{*2}$, then $F(\hat\gamma;\tau^*) < F(1;\tau^*)$, and with Theorem 1 every minimizer of $F(\cdot\,;\tau^*)$ is interior.*

Proof. (i) In (2) the cross-block enters only through $\gamma\widehat{B}D^{-1}$ and $\gamma\widehat{B}D^{-1}\widehat{B}^\top$, so every occurrence of the perturbation τN_B carries the prefactor γ, and the deeper levels operate on augmented blocks that retain it. Hence $\partial w/\partial\tau = \gamma\, h(\gamma,\tau)$ with h bounded on the stated compact set, so $V(\gamma;\tau N) - V(\gamma;0)$ is Lipschitz in τ with constant $O(\gamma)$; by symmetry of N the first-order term in τ cancels in expectation, and Taylor's theorem with the uniformly bounded second derivative gives (i). (ii) is arithmetic: $F(\hat\gamma;\tau^*) \le V_0(\hat\gamma) + C\hat\gamma^2\tau^{*2} < F(1;\tau^*)$. □

Remark 2. The sign of $F - V_0$ is not fixed: symmetric noise can reduce expected out-of-sample variance slightly at moderate γ through the curvature of V in the perturbation (observed in the witness family). The theorems bound magnitudes; what they use is only that noise costs nothing at $\gamma = 0$ and that its magnitude at $\gamma = 1$ can exceed the population gain of full coupling — dramatically so once the perturbed recursion approaches the boundary of the positive-definite cone.

Proposition 2 (Exact witness)**.** *Let $n = 4$ with blocks $\{1,2\},\{3,4\}$, volatilities $(1,1,2,2)$, within-block correlation $\frac{1}{2}$, cross-block correlation $\frac{3}{10}$; let the estimate err in the $(1,3)$ covariance entry by $\pm\frac{3}{20}\cdot 2$ with probability $\frac{1}{2}$ each. Then, in exact rational arithmetic,*

$$F(\tfrac{9}{10}) < F(0) \quad (\textit{margin } 4.879\times 10^{-2}), \qquad F(\tfrac{9}{10}) < F(1) \quad (\textit{margin } 5.702\times 10^{-4}),$$

so every minimizer of F on $[0,1]$ is strictly interior. All augmented blocks remain positive definite on both noise branches for every $\gamma \in [0,1]$: no repair or projection is involved. This is the minimal non-trivial configuration: with singleton-children nodes performing no augmentation, $n \le 3$ admits no γ-dependence at all, so $n = 4$ in two blocks of two is the smallest universe on which the bridge exists.

Proof. F at rational arguments is an exact rational number, being a finite composition of field operations (2) on rational inputs. The two inequalities are evaluated in exact fraction arithmetic by the accompanying certificate script (see the title footnote), which also certifies positive definiteness along the path. □

4 The bridge as a bias–variance frontier

Smoothness gives the uniform second-order expansion

$$F(\gamma;\tau) \;=\; V_0(\gamma) + \tau^2 G(\gamma) + O(\tau^4), \qquad G(0) = G'(0) = 0, \tag{5}$$

the boundary behavior of the noise-cost profile G coming from Lemma 1 and the γ-prefactor of Theorem 2(i). Write $F_2 = V_0 + \tau^2 G$.

Theorem 3 (The optimal coupling sweeps the bridge monotonically)**.** *Suppose on* $(0, \bar{\gamma})$ *that* $V_0' < 0$, $V_0'' \geq 0$ *(decreasing convex population curve) and* $G' > 0$, $G'' \geq 0$ *(increasing convex noise cost), with at least one of* V_0'', G'' *strictly positive. Then* $t(\gamma) = -V_0'(\gamma)/G'(\gamma)$ *is a continuous, strictly decreasing bijection onto its range, and for every* τ^2 *in that range the surrogate* $F_2(\cdot\,; \tau)$ *has the unique minimizer* $\gamma^*(\tau^2) = t^{-1}(\tau^2)$*, strictly decreasing in the noise level. Within the surrogate:* $\gamma^* \to \bar{\gamma}$ *only as the noise vanishes,* $\gamma^* \to 0$ *only as it diverges, and every interior coupling is optimal at exactly one noise level.*

Proof. $t' = (-V_0''G' + V_0'G'')/G'^2 < 0$ on the stated domain, so t is a strictly decreasing bijection; $F_2' = V_0' + \tau^2 G'$ is negative before $t^{-1}(\tau^2)$ and positive after. The comparative statics $d\gamma^*/d\tau^2 = -G'/F_2'' < 0$ follow from the implicit function theorem. □

Proposition 3 (Transfer to the exact objective)**.** *Suppose in addition* $F_2'' \geq c > 0$ *on a neighborhood of* $\gamma^*(\tau^2)$ *and* $|F - F_2| \leq K\tau^4$ *there. Then any minimizer of* F *in that neighborhood satisfies* $|\gamma_F - \gamma^*| \leq 2\sqrt{K/c}\,\tau^2$*.*

Proof. $\frac{c}{2}(\gamma_F - \gamma^*)^2 \leq F_2(\gamma_F) - F_2(\gamma^*) \leq \left[F(\gamma_F) - F(\gamma^*)\right] + 2K\tau^4 \leq 2K\tau^4$. □

Remark 3 (Interpretation)*.* γ is a shrinkage intensity toward the noise-immune end of the bridge, and Theorem 3 is an optimal-shrinkage formula in the spirit of [7]: the first-order condition balances the marginal population value of coupling against its marginal noise cost. The endpoints recover the prior results: as $\gamma \to 1$ the augmentation approaches the exact Schur complement and G' inherits the inversion sensitivity of minimum variance — the instability quantified at the endpoint by [5] — so full coupling is optimal only as the noise vanishes, exactness [6] notwithstanding; as $\gamma \to 0$, G' vanishes while the population gain $-V_0'(0)$ does not.

Remark 4 (Verification in the witness family)*.* With symmetric two-point noise on the cross-block correlation of the Proposition 2 family, the hypotheses hold: V_0 is decreasing convex on $[0, 1)$; G is increasing convex, vanishing quadratically at 0 ($G \approx 0.003, 0.032, 0.138, 0.448$ at $\gamma = 0.2, 0.4, 0.6, 0.8$), τ-stable for $\tau \leq 0.2$. The root of $V_0' + \tau^2 G' = 0$ tracks the exact minimizer of F:

τ	0.10	0.15	0.20	0.25	0.30	0.35
exact $\arg\min F$	0.85	0.70	0.65	0.55	0.50	0.45
root of $V_0' + \tau^2 G'$	0.83	0.74	0.66	0.59	0.53	0.49

(grid resolution 0.05): the optimal coupling falls monotonically from near full coupling toward the middle of the bridge as the noise grows, as Theorem 3 requires.

5 Closing

Exactness of full coupling does not confer statistical optimality, and the noise-robustness of HRP does not confer it either: under covariance estimation error the Schur coupling is a bias–variance dial whose optimal setting is interior, decreasing in the noise, and characterized by $\tau^2 = -V_0'/G'$. The empirical counterpart — a walk-forward bridge on Dow constituents with one-year training windows, minimum near $\gamma \approx 0.55$ and robust to Ledoit–Wolf shrinkage of the inputs — is interactive at `schur.microprediction.org` (the “Demo” page), with the wider demonstration suite at `allocation.microprediction.org`.

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